

Challenge 20 – AI-Powered Personalized Learning & Teacher Assistant Platform

Domain: Education

The Challenge

Many government and rural schools in Gujarat face challenges such as diverse student learning levels, limited access to personalized learning resources, increasing administrative workload for teachers, and insufficient real-time insights into student progress. Teachers require AI-powered assistance to create engaging lesson plans, assess student performance, and provide individualized learning support while reducing repetitive tasks.

Objective

Create Agentic AI solutions that personalize student learning experiences, assist teachers in lesson planning and content generation, automate assessment and feedback, identify students requiring additional support, and provide actionable insights to improve learning outcomes.

Suggested AI Agents

- Personalized Learning Recommendation Agent
- AI Lesson Plan & Content Generation Agent
- Student Performance & Learning Analytics Agent
- Automated Assessment & Feedback Agent
- Teacher Assistant & Classroom Support Agent
- IBM SkillsBuild learning resources
- Open Educational Resources (OER)

Technology to be used - IBM Bob + IBM Granite LLM + IBM Cloud